

QUESTION 1 (25 MARKS)

The failure of the LAS CAD implementation in 1992 was affected by several dependability issues.		
1.1	Discuss the dependability shortcomings in this system on the following dimensions – reliability, availability, safety, security and usability. Support your explanations with suitable examples.	10
1.2	Explain how these shortcomings could be addressed if this system were to be developed today using modern technologies, techniques, approaches or processes.	10
1.3	Would it be easier to measure and quantify the reliability of this system or its usability? Explain why this is so.	5

QUESTION 2 (25 MARKS)

The *"incomplete and untested"* status of the LAS CAD system was seen as an important contributor to the system's failure. Discuss how you would avoid this during the testing phase in the following scenarios -

- 2.1 Assume that a mobileApp has been developed for the **LAS CAD** system and your team is responsible for conducting the conceptual testing, user experience testing, and security testing of the app. Explain what you would be testing in each case as well as two techniques that could assist you with this. 15
- 2.2 According to the report a key lesson learnt from the failure of the LAS CAD system was that – 10

"The Development process must allow fully for consultation, quality assurance and testing".

Explain how you would ensure that these qualities are met when developing the mobileApp using an Agile approach.

QUESTION 3 (25 MARKS)

- 3.1 Large projects like the **LAS CAD** system require careful management tracking and control for it to succeed. Discuss five umbrella activities which would accompany the software development process. 15
- 3.2 People are the most important element of a successful project. Identify five stakeholders involved in the **LAS CAD** system and explain their role in the project. 10

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